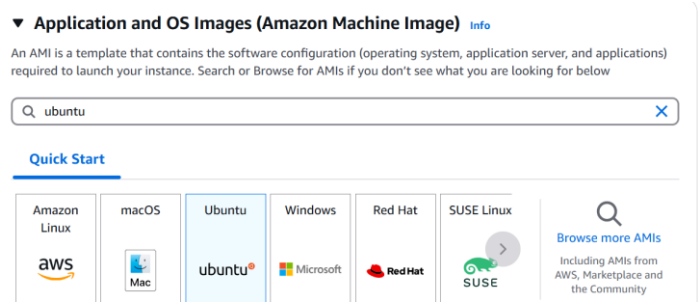
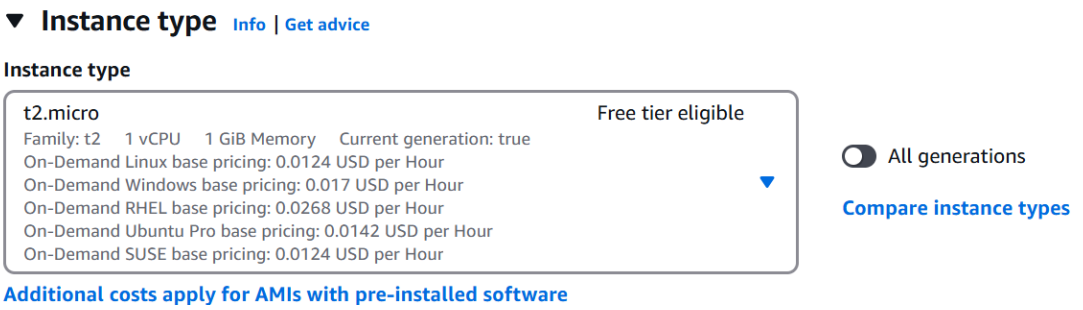
Open aws account
Select EC2 instance
Click on launch instance
Select ubuntu os

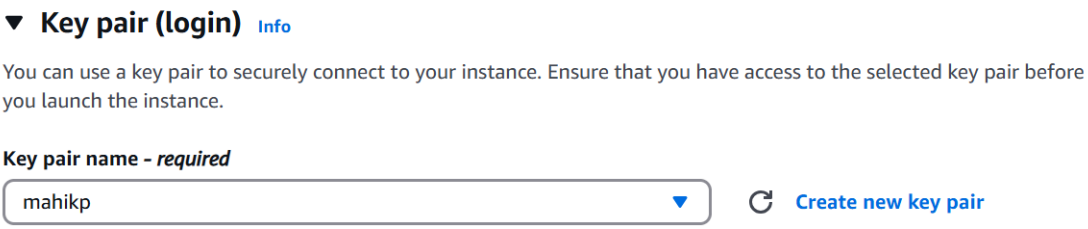


Select ubuntu free tier as amazoin image

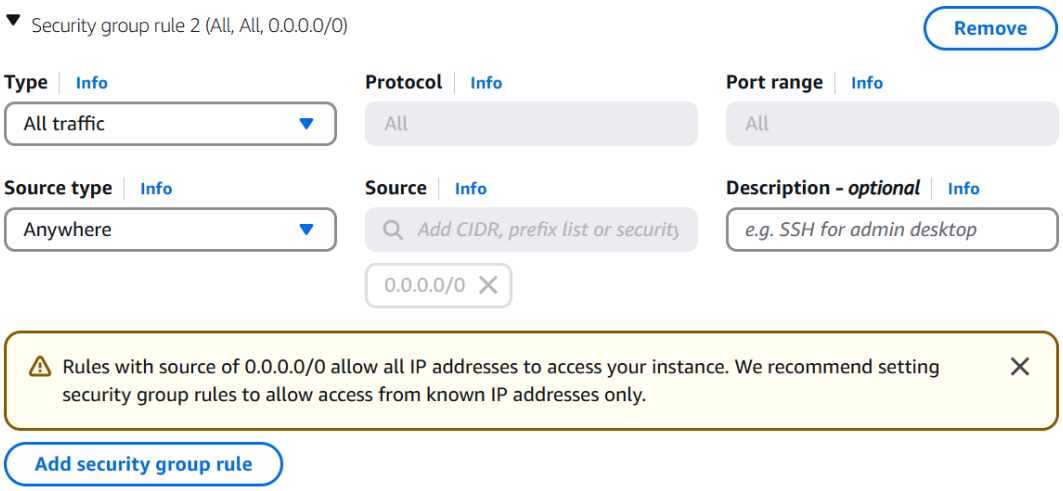
Select instance type t2.micro



Create key pair

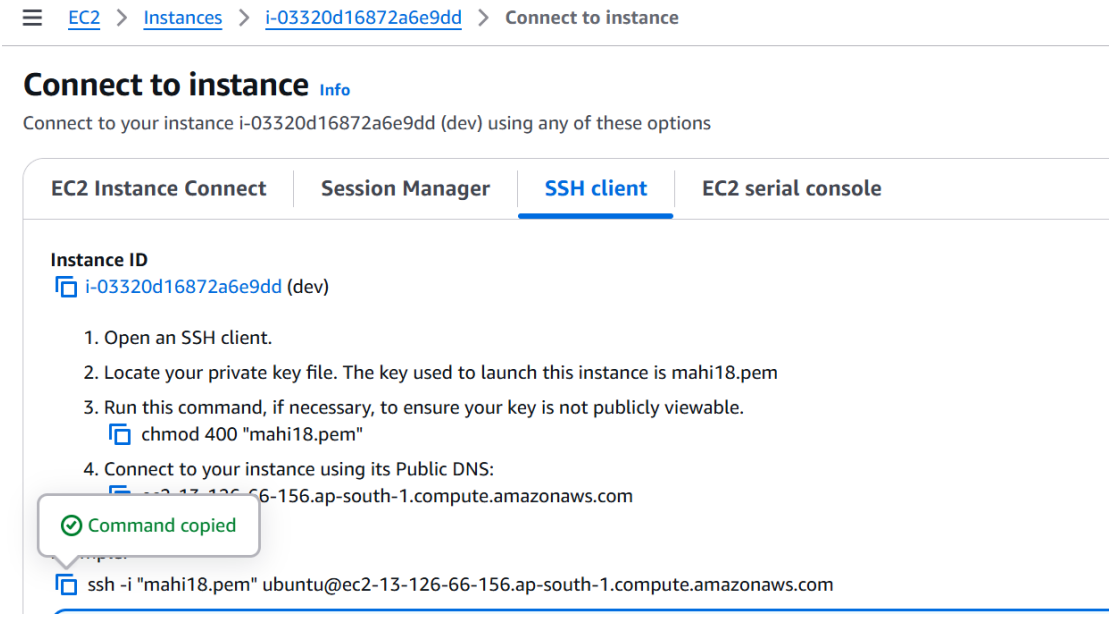


Network add the below rule



Goto the .pem file location from there open gitbash (before you need to download and install gitbash in your computer)

Goto aws click on the connect take the ssh id copy and paste in the gitbash




```
Admin@DESKTOP-ASGG5M3 MINGW64 ~/Downloads
$ ssh -i "mahi18.pem" ubuntu@ec2-13-126-66-156.ap-south-1.compute.amazonaws.com
The authenticity of host 'ec2-13-126-66-156.ap-south-1.compute.amazonaws.com (13.126.66.156)' can't be established.
ED25519 key fingerprint is SHA256:UvyPb6zcAYpoJGBzL+spFknaVb2FEEqNfBaIC0LB08M.
This key is not known by any other names.
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
```

Now u need to install jenkins

First u need to update the apt repository

$ sudo apt update

```
ubuntu@ip-172-31-14-226:~$ sudo apt update
Hit:1 http://ap-south-1.ec2.archive.ubuntu.com/ubuntu noble InRelease
Get:2 http://ap-south-1.ec2.archive.ubuntu.com/ubuntu noble-updates InRelease [126 kB]
Get:3 http://ap-south-1.ec2.archive.ubuntu.com/ubuntu noble-backports InRelease [126 kB]
Get:4 http://ap-south-1.ec2.archive.ubuntu.com/ubuntu noble/universe amd64 Packages [15.0 MB]
Get:5 http://security.ubuntu.com/ubuntu noble-security InRelease [126 kB]
Get:6 http://ap-south-1.ec2.archive.ubuntu.com/ubuntu noble/universe Translation-en [5982 kB]
```

Install java

sudo apt install openjdk-21-jdk -y

```
Running kernel seems to be up-to-date.

No services need to be restarted.

No containers need to be restarted.

No user sessions are running outdated binaries.

No VM guests are running outdated hypervisor (qemu) binaries on this host.
ubuntu@ip-172-31-14-226:~$ java --version
openjdk 21.0.5 2024-10-15
OpenJDK Runtime Environment (build 21.0.5+11-Ubuntu-1ubuntu124.04)
OpenJDK 64-Bit Server VM (build 21.0.5+11-Ubuntu-1ubuntu124.04, mixed mode, sharing)
ubuntu@ip-172-31-14-226:~$
```

u need to install git and maven and check it
Sudo apt-get install -y git maven
git --version
mvn --version

```
Setting up liberror-prone-java (2.18.0-1) ...
Setting up libguice-java (4.2.3-2) ...
Setting up libmaven3-core-java (3.8.7-2) ...
Setting up maven (3.8.7-2) ...
update-alternatives: using /usr/share/maven/bin/mvn to provide /usr/bin/mvn (mvn) in auto mode
Scanning processes...
Scanning linux images...

Running kernel seems to be up-to-date.

No services need to be restarted.

No containers need to be restarted.

No user sessions are running outdated binaries.

No VM guests are running outdated hypervisor (qemu) binaries on this host.
ubuntu@ip-172-31-14-226:~$ git --version
git version 2.43.0
ubuntu@ip-172-31-14-226:~$ mvn --version
Apache Maven 3.8.7
Maven home: /usr/share/maven
Java version: 21.0.5, vendor: Ubuntu, runtime: /usr/lib/jvm/java-21-openjdk-amd64
Default locale: en, platform encoding: UTF-8
OS name: "linux", version: "6.8.0-1021-aws", arch: "amd64", family: "unix"
ubuntu@ip-172-31-14-226:~$
```

Open jenkin.io (Download and deploy) in browser and select the generic java package(.war)

Copy and paste link addrest

$wget linkaddress

```
ubuntu@ip-172-31-14-226:~$ wget https://get.jenkins.io/war-stable/2.479.3/jenkins.war
--2025-01-21 02:09:50--  https://get.jenkins.io/war-stable/2.479.3/jenkins.war
Resolving get.jenkins.io (get.jenkins.io)... 20.7.178.24, 2603:1030:408:5::15a
Connecting to get.jenkins.io (get.jenkins.io)|20.7.178.24|:443... connected.
HTTP request sent, awaiting response... 302 Found
Location: https://2.mirrors.in.sahilister.net/jenkins/war-stable/2.479.3/jenkins.war [following]
--2025-01-21 02:09:51--  https://2.mirrors.in.sahilister.net/jenkins/war-stable/2.479.3/jenkins.war
Resolving 2.mirrors.in.sahilister.net (2.mirrors.in.sahilister.net)... 82.180.146.159, 2400:d321:2217:1599::1
Connecting to 2.mirrors.in.sahilister.net (2.mirrors.in.sahilister.net)|82.180.146.159|:443... connected.
HTTP request sent, awaiting response... 200 OK
Length: 96806970 (92M) [application/java-archive]
Saving to: 'jenkins.war'

jenkins.war                         100%[===================================================================================>]  92.32M  8.67MB/s    in 10s

2025-01-21 02:10:02 (9.08 MB/s) - 'jenkins.war' saved [96806970/96806970]

ubuntu@ip-172-31-14-226:~$
```

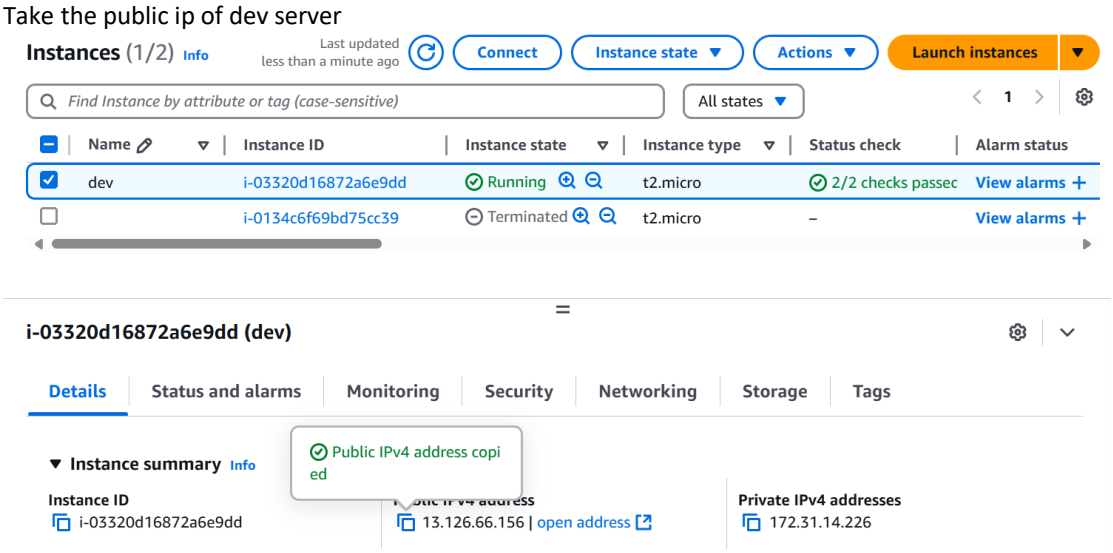
Start the jenkins services

$ java -jar jenkins.war

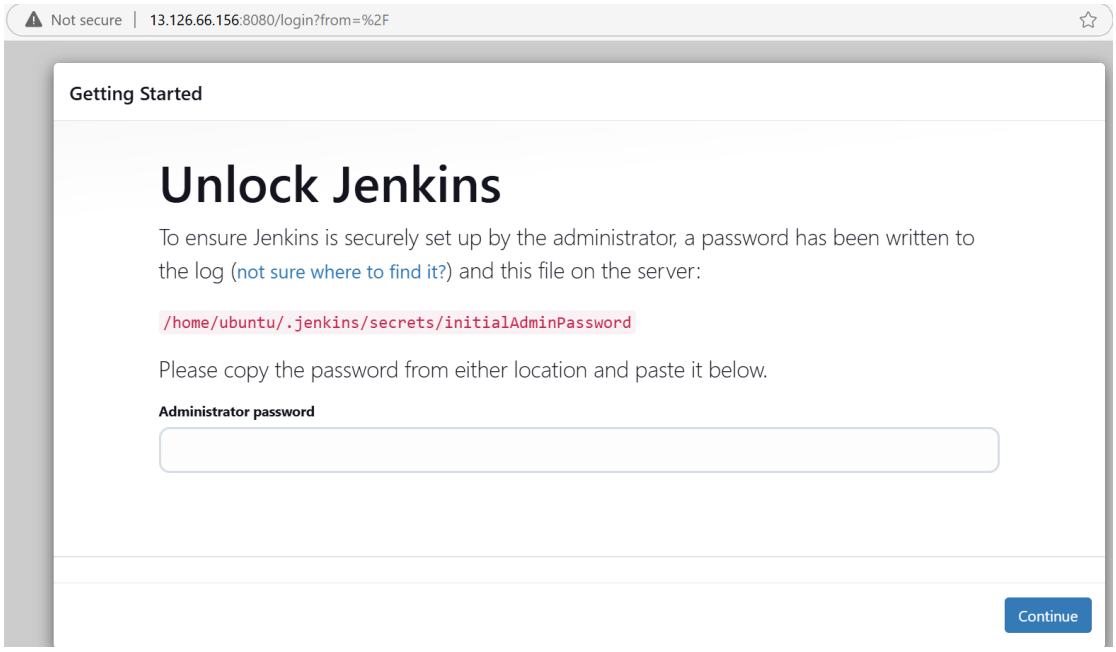
```
2025-01-21 02:11:18.931+0000 [id=31]    INFO    jenkins.InitReactorRunner$1#onAttained: Started all plugins
2025-01-21 02:11:18.943+0000 [id=32]    INFO    jenkins.InitReactorRunner$1#onAttained: Augmented all extensions
2025-01-21 02:11:19.329+0000 [id=32]    INFO    jenkins.InitReactorRunner$1#onAttained: System config loaded
2025-01-21 02:11:19.330+0000 [id=32]    INFO    jenkins.InitReactorRunner$1#onAttained: System config adapted
2025-01-21 02:11:19.330+0000 [id=32]    INFO    jenkins.InitReactorRunner$1#onAttained: Loaded all jobs
2025-01-21 02:11:19.331+0000 [id=32]    INFO    jenkins.InitReactorRunner$1#onAttained: Configuration for all jobs updated
2025-01-21 02:11:19.380+0000 [id=45]    INFO    hudson.util.Retrier#start: Attempt #1 to do the action check updates server
2025-01-21 02:11:20.359+0000 [id=32]    INFO    jenkins.install.SetupWizard#init:

*************************************************************
*************************************************************
*************************************************************

Jenkins initial setup is required. An admin user has been created and a password generated.
Please use the following password to proceed to installation:

5103d95754da497eac50d479875e20d9

This may also be found at: /home/ubuntu/.jenkins/secrets/initialAdminPassword

*************************************************************
*************************************************************
*************************************************************

2025-01-21 02:11:27.629+0000 [id=31]    INFO    jenkins.InitReactorRunner$1#onAttained: Completed initialization
2025-01-21 02:11:27.662+0000 [id=24]    INFO    hudson.lifecycle.Lifecycle#onReady: Jenkins is fully up and running
2025-01-21 02:11:29.272+0000 [id=45]    INFO    h.m.DownloadService$Downloadable#load: Obtained the updated data file for hudson.tasks.Maven.MavenInstaller
2025-01-21 02:11:29.274+0000 [id=45]    INFO    hudson.util.Retrier#start: Performed the action check updates server successfully at the attempt #1
```

How can we access the jenkin

Take the public ip of dev server



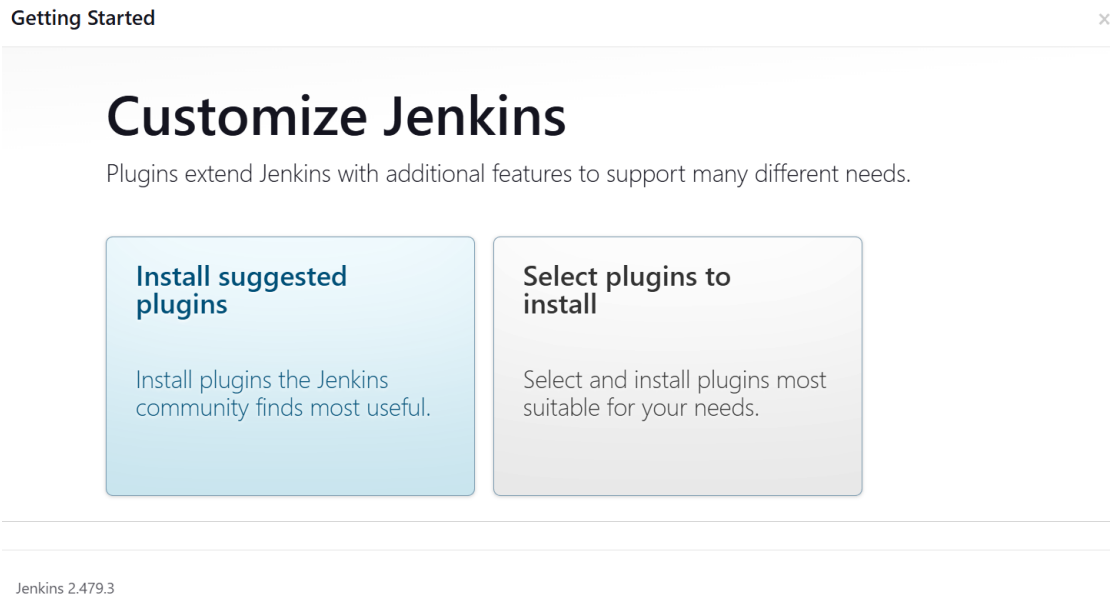
Copy and paste in the browser add 8080 as port number



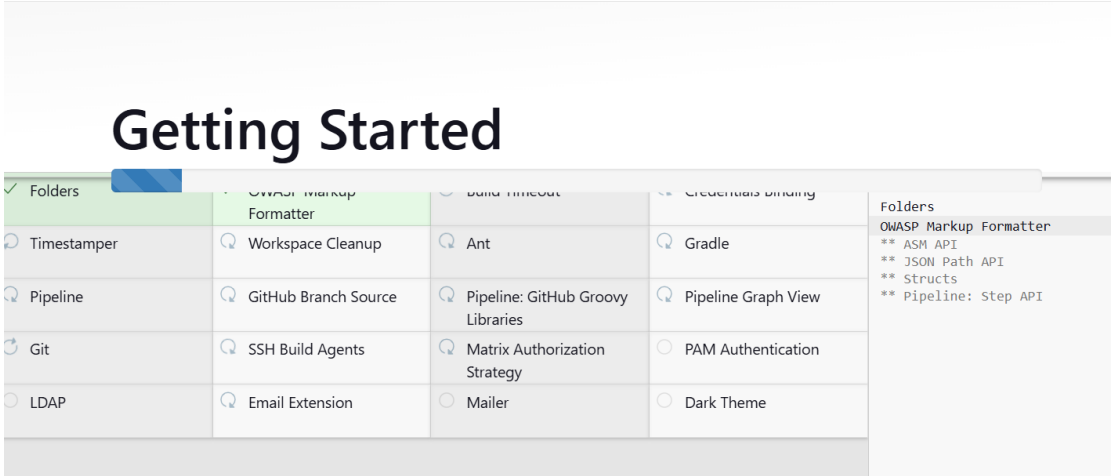
Unlock the jenkins with password which is present in the gitbash

```
Jenkins initial setup is required. An admin user has been created and a password generated.
Please use the following password to proceed to installation:

5103d95754da497eac50d479875e20d9

This may also be found at: /home/ubuntu/.jenkins/secrets/initialAdminPassword
```

Select the install suggested pluggins



Getting Started

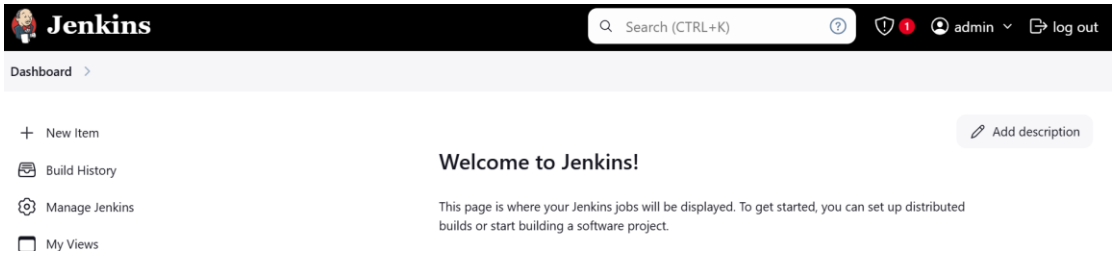


You can give all admin

Click on save and continue

Click on save and finish

Your jenkins is ready



Note: don’t close the jenkins server